Phishing URL Detection Using Machine Learning

Final Project Report - Track 1 / Problem 02

Dataset note: The included dataset is synthetic and was created for an educational prototype. It is not an externally sourced real-world phishing dataset.

Abstract

This project develops a lightweight machine-learning prototype for classifying URLs as phishing or legitimate. The workflow covers data loading, cleaning, exploratory analysis, feature selection, model training, evaluation and a Streamlit demonstration application.

1. Introduction

Phishing websites imitate trusted services to deceive users. URL structure provides measurable signals such as length, domain structure, special-character usage and protocol information. The project studies whether foundational machine-learning models can separate the two classes in a small educational dataset.

2. Problem Definition

Given URL-derived structural features, predict whether a URL belongs to the phishing class or the legitimate class.

3. Objective

Build a reproducible, zero-cost prototype that prepares the data, compares foundational classifiers, evaluates the results and demonstrates prediction through Streamlit.

4. Dataset

The project uses 3,000 synthetic records: 1,500 phishing and 1,500 legitimate. The features are URL, URLLength, SubdomainCount, SpecialCharCount, DigitRatio, DomainLength, HTTPS, HasIP, URLDepth, HyphenCount, DotCount, and Label. The balanced class distribution simplifies the educational demonstration.

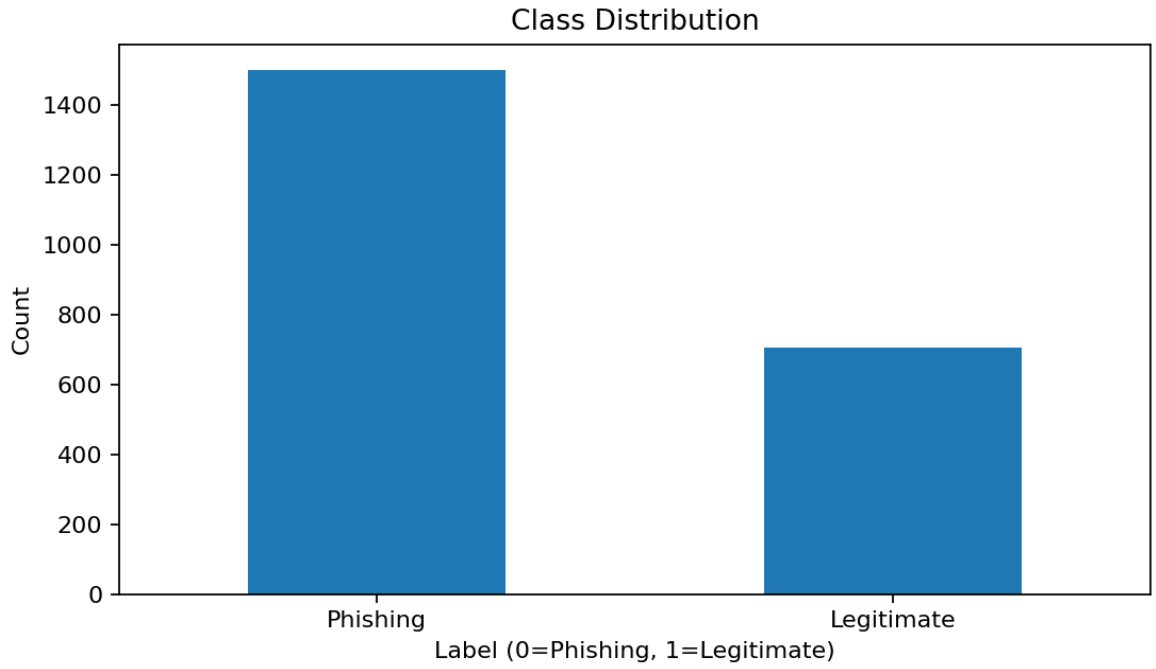


Figure 1. Class distribution of the synthetic dataset.

5. Data Preprocessing

The notebook checks data types, missing values and duplicate rows. Duplicate records are removed. The URL text is not directly fed to the initial baseline models; instead, numerical URL-structure features are used.

6. Exploratory Data Analysis

EDA includes class distribution, URL-length distribution and HTTPS-vs-class analysis. These plots help identify differences between phishing and legitimate records and verify that the dataset is suitable for classification.

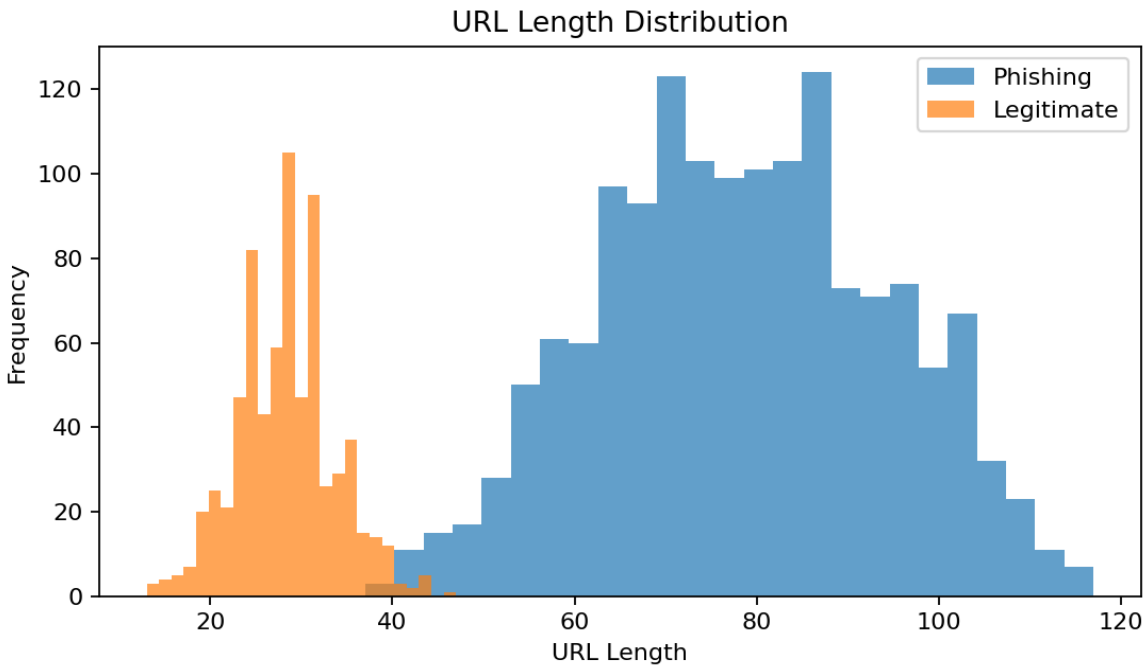


Figure 2. URL-length distribution.

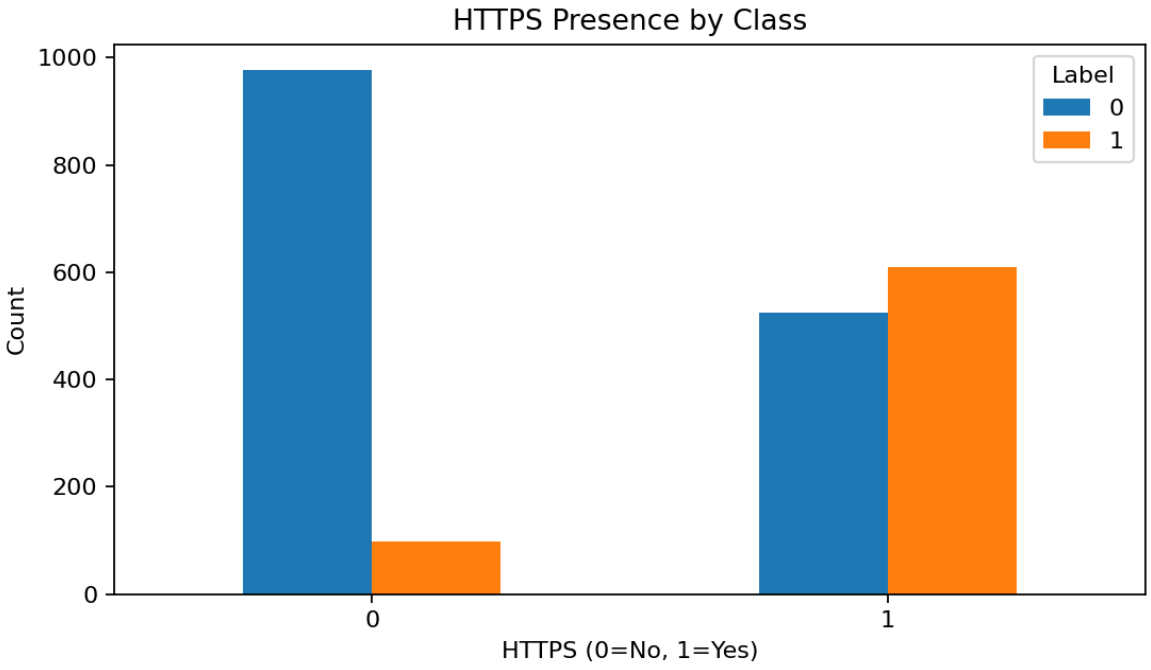


Figure 3. HTTPS presence by class.

7. Methodology

The data is split into 80% training and 20% test data using stratification. Decision Tree and Logistic Regression are evaluated. The Decision Tree is retained for the prototype because it is easy to explain and does not require feature scaling for this feature set.

8. Results

Both baseline models reach perfect metrics on the held-out test split of the synthetic dataset. This should be interpreted as a property of the constructed prototype dataset, not as evidence that phishing can be detected with perfect accuracy in real-world traffic.

Model	Accuracy	Precision	Recall	F1-score
Decision Tree	1.00	1.00	1.00	1.00
Logistic Regression	1.00	1.00	1.00	1.00

9. Prototype

The Streamlit application accepts a URL, extracts the same numerical features used during model training and displays either PHISHING or LEGITIMATE together with the extracted features.

10. Limitations

The dataset is synthetic, balanced and relatively simple. The model is therefore not validated against diverse, changing real-world URLs. URL parsing is intentionally lightweight, and the prototype should be treated as an educational demonstration rather than a security product.

11. Future Scope

Future work can replace the synthetic dataset with a credited public dataset, add lexical and host-based signals, evaluate cross-dataset generalization, tune decision thresholds and add stronger security-oriented validation.

12. Conclusion

The project demonstrates an end-to-end phishing URL classification workflow using classical machine learning and a Streamlit interface. It satisfies the requested educational pipeline while making the dataset limitation explicit.

References / Credit: Supplied capstone problem statement; project-generated synthetic dataset; scikit-learn, pandas and Streamlit documentation used for implementation.